

FMUSER FUQ-200F QAM Processor Multiplexer Scrambler Modulator 1024 IP Input DVB-C 192 QAM Output 10G-1G SFP+ Ports CAS Encryption for Cable TV Broadcasting Product Overview



The FUQ-200F integrates multiplexer, scrambler, and modulator in a 1U chassis, with IP inputs via SFP+, SFP, and GE ports to output up to 192 DVB-C QAM carriers. It processes IP streams using CAS encryption and PSI/SI editing for signal modulation. This modular solution features QAM cards (32/48/64ch) and web-based control for easy operation. Ideal for broadcast headend and cable TV systems requiring high-density signal processing. This processor is equipped with 3 card slots to load 32ch or 48ch or 64ch QAM card. Its high density helps operators to save cost to the most extent.

Key feature

Supports modular QAM cards (32/48/64ch) in 1U for high-density output.

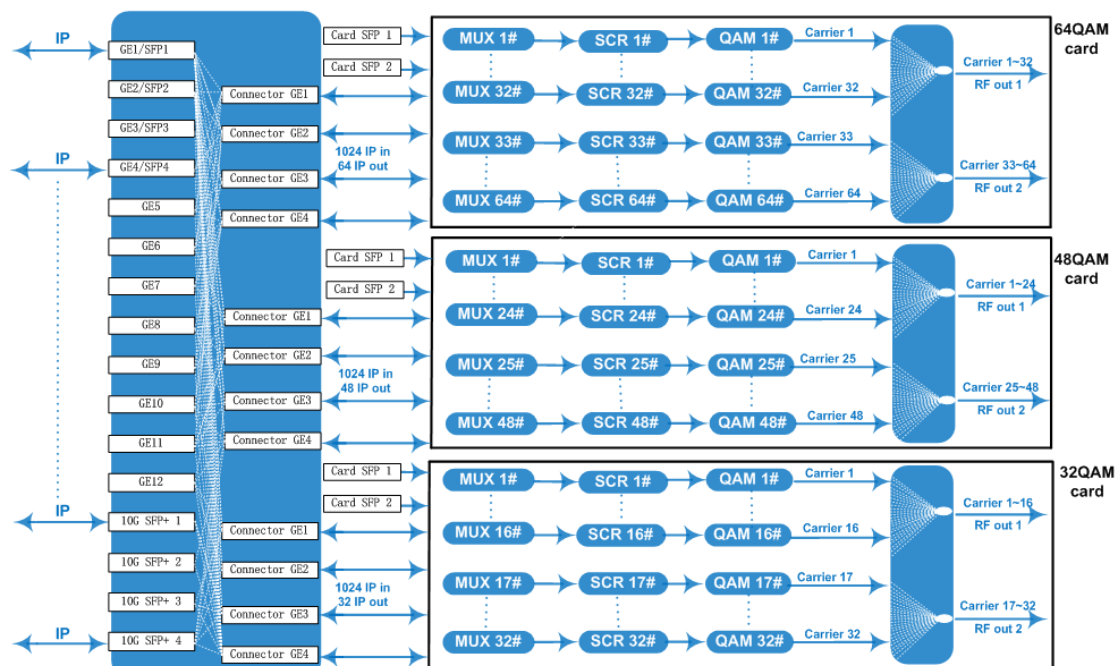
Features IP inputs via 4× SFP+ (10G), 4× SFP (1G), and 12× GE ports.

Enables multiplexer/scrambler functions with up to 6 CAS encryption standards.

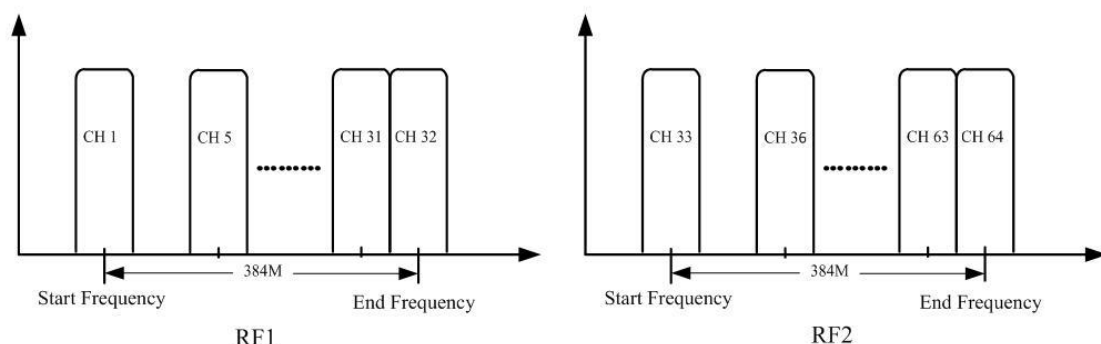
Generates 192 non-adjacent DVB-C QAM carriers via J.83A/B/C modulation.

Provides web-based control for RF output (-20dBm~+10dBm) and symbol rates (3600-7200Ksps).

Inner Principle Chart (Example: one 64qam card + one 48qam card + one 32qam card)



Carrier Setting Illustration (Example: 64QAM card)



Specifications

Input	IP Input	Max1024 IP inputs per card through 4 SFP+, 4 SFP and 12 GE ports in the front panel and 2 SFP ports in the card itself	Max1024 IP inputs per card through 4 SFP+, 4 SFP and 12 GE ports in the front panel and 2 SFP ports in the card itself
Input	Transmission Rate	Max 9600Mbps for each 10G SFP+ input/Max 960Mbps for each GE and SFP input	Max 9600Mbps for each 10G SFP+ input/Max 960Mbps for each GE and SFP input

Input	Transport Protocol	TS over UDP/RTP, unicast and multicast, IGMP V2/V3	TS over UDP/RTP, unicast and multicast, IGMP V2/V3
Mux	Max PIDs Remapping	256 per output channel	256 per output channel
Mux	Functions	PID remapping (auto/manually optional)	PID remapping (auto/manually optional)
Mux	Functions	PCR accurate adjusting	PCR accurate adjusting
Mux	Functions	PSI/SI table automatically generating	PSI/SI table automatically generating
Scrambling	Max simulcrypt CA	6	6
Scrambling	Scramble Standard	ETR289, ETSI 101 197, ETSI 103 197	ETR289, ETSI 101 197, ETSI 103 197
Scrambling	Connection	Local/remote connection	Local/remote connection
Modulation	Modulation Standard	EN300 429/ITU-T J.83A/B/C	EN300 429/ITU-T J.83A/B/C
Modulation	Constellation	J.83A	Constellation :16/32/64/128/256QAM
Modulation	Constellation	J.83A	Bandwidth :8M
Modulation	Constellation	J.83B/C	Constellation :64/256QAM
Modulation	Constellation	J.83B/C	Bandwidth :6M
Modulation	QAM Channel	32/48/64/80/96/112/128/144/160/176/192 non-adjacent carriers output, 384Mbps bandwidth for each RF port	32/48/64/80/96/112/128/144/160/176/192 non-adjacent carriers output, 384Mbps bandwidth for each RF port
Modulation	Symbol Rate	3600~7200Ksps, 1ksps stepping 5057Ksps (J.83B, 64QAM) ; 5361Ksps (J.83B, 256QAM)	3600~7200Ksps, 1ksps stepping 5057Ksps (J.83B, 64QAM) ; 5361Ksps (J.83B, 256QAM)
Modulation	Constellation	16, 32, 64, 128, 256QAM	16, 32, 64, 128, 256QAM
Modulation	FEC	RS (204, 188)	RS (204, 188)

RF Output	Interface (per card)	2 F type output ports for 32/48/64 carriers card, 75Ω. 32ch/48ch/64ch qam card : Carrier 1~16/1~24/1~32 out thru RF1, 17~32/25~48/33~64 thru RF2	2 F type output ports for 32/48/64 carriers card, 75Ω. 32ch/48ch/64ch qam card : Carrier 1~16/1~24/1~32 out thru RF1, 17~32/25~48/33~64 thru RF2
RF Output	RF Range	50~960MHz, 1kHz stepping	50~960MHz, 1kHz stepping
RF Output	Output Level	-20dBm~+10dBm(87~117dBμV), 0.1dB stepping	-20dBm~+10dBm(87~117dBμV), 0.1dB stepping
RF Output	MER	≥ 40dB	≥ 40dB
TS output	Per Daughter QAM Card	32/48/64 IP output over UDP/RTP/RTSP, unicast/multicast, through 4 SFP+ ports, 4 SFP ports and 12 GE ports in the front panel	32/48/64 IP output over UDP/RTP/RTSP, unicast/multicast, through 4 SFP+ ports, 4 SFP ports and 12 GE ports in the front panel
System	Web-based Network management	Web-based Network management	Web-based Network management
General	Dimension	480mm×440mm×44.5 mm (W×L×H)	480mm×440mm×44.5 mm (W×L×H)
General	Temperature	0~45℃(operation), -20~80℃(storage)	0~45℃(operation), -20~80℃(storage)
General	Power Supply	AC 100V±10%, 50/60Hz ; or AC 220V±10%, 50/60Hz	AC 100V±10%, 50/60Hz ; or AC 220V±10%, 50/60Hz
General	Consumption	About 200W (3 daughter QAM cards)	About 200W (3 daughter QAM cards)

Major Technical Comparison

	DHP200D	DHP200E
Card Number	Max 2	Max 3
SFP+/SFP/RJ45 ports in the front panel	3 SFP+(10G)/SFP(1G) ports and 2 GE ports	4 SFP+ (10G)/4 SFP(1G) ports and 12 GE ports

Max Carriers number	112	192
Inner connector GE ports	6	12
DATA Port in card	DepeFU on card spec	2 SFP ports as input per card

Order Guide

Card Type Option: 32, 48 or 64 QAM card

Card Number Option: 1, 2 or 3 pcs

SFP ports option in card: With or Without

PSU option: 1*220VAC or 2*220VAC not hot-swapping PSU

About Us

FMUSER is a company specializing in broadcasting and television transmission systems and integrating hotel IPTV systems. The company was established in 2009 and has served over 180 countries worldwide. More than 20,000 television stations, radio stations, and hotel clients from different.

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